# Project Report: RANDOM FOREST MACHINE LEARNING PROJECT

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**1. Project Title:**
Random Forest Machine Learning Project – Car Evaluation

**2. Dataset Name:**
Car Evaluation Dataset (car_evaluation.csv), obtained from "archive.ics.uci.edu". The dataset contains six input features: buying price, maintenance cost, number of doors, passenger capacity, luggage boot size, and safety. The target variable is the car evaluation category.

Dataset Link: https://archive.ics.uci.edu/dataset/19/car+evaluation

**3. Problem Statement:**
The objective of this project is to develop a Machine Learning model using the Random Forest algorithm to predict the acceptability of a car based on its features. The model classifies cars into four categories: Unacceptable, Acceptable, Good, and Very Good.

**4. Data Preprocessing Steps:**

- Loaded the dataset using Pandas.
- Assigned appropriate column names to the dataset.
- Performed data exploration and checked for missing values.
- Handled missing values using the most frequent value.
- Converted categorical variables into numerical format using One-Hot Encoding.
- Separated independent variables (X) and dependent variable (y).
- Split the dataset into 80% training data and 20% testing data.

**5. Model Used:**
Random Forest Classifier was used with 300 decision trees and balanced class weights. The model combines multiple decision trees to improve prediction accuracy and reduce overfitting.

**6. Evaluation Metrics:**
The model was evaluated using the following metrics:

- Accuracy: 97.40%
- Precision (Weighted): 97.46%
- Recall (Weighted): 97.40%
- F1-Score (Weighted): 97.41%
- Confusion Matrix: Used to analyze correct and incorrect predictions across all four classes.

**7. Key Findings:**

- The Random Forest model achieved strong classification performance.
- The model successfully classified cars into four evaluation categories.
- Feature importance analysis helped identify the most influential attributes in predicting car acceptability.
- One-Hot Encoding enabled the model to process categorical data effectively.
- The model demonstrated good predictive performance on the test dataset.

**8. Conclusion:**

The project successfully demonstrates the implementation of the Random Forest algorithm on a real-world Car Evaluation dataset. The model achieved 97.40% accuracy, showing its effectiveness in predicting car acceptability. This project provided practical knowledge of data preprocessing, categorical encoding, model training, evaluation metrics, and feature importance. Further improvements can be achieved through hyperparameter tuning and cross-validation.